

1 OBJECTIVE

Ph.D. applicant focusing on deployable and reliable perception for real-world environments, particularly medical settings. Current research centers on **spatial perception**, **Vision-language-action models**, **World models**, and **Safety-aware embodied AI**.

2 EDUCATION

Kennesaw State University, M.S. Software Engineering

2025.1 – Present

Graduate Research Assistant

Heilongjiang University, B.Eng. Computer Network Engineering

2016.9 – 2020.7

National Encouragement Scholarship; First-Class Scholarship (2018, 2019)

3 RESEARCH EXPERIENCE

Synthetic-First Data Engine for Medical Perception (Paper in prep)

Project Homepage

- Built a **hierarchical perception prototype** for **deltoid injection guidance**, integrating **synthetic data generation**, **segmentation**, **geometric keypoint extraction**, and **safety-zone reasoning**.
- **Developed dataset curation tools** and multi-view generation templates, along with pseudo-label refinement modules to improve anatomical boundary precision.
- Investigating **synthetic-to-real transfer** using segmentation metrics, boundary distance, and medically grounded safety indicators.

Safety-Aware Embodied AI for Real-World Task Assistance (Research Direction)

- Investigating perception pipelines that connect segmentation, geometric reasoning, and spatial understanding for embodied decision-making.
- Exploring **safety-aware perception** and interpretable sensing and reasoning strategies for deployable medical assistant agents.
- Investigating **agent-centric learning frameworks** that integrate data generation, reinforcement learning, and human-in-the-loop feedback for continual adaptation in embodied systems.

4 HONORS & AWARDS

Second Place, KSU C-Day Computing Showcase (Fall 2025).

5 WORK EXPERIENCE

City University of Hong Kong, Research Assistant

2024.11 – 2025.1

Tech: Python, React

- Developed a secure, encrypted data-marketing platform supporting cross-institution research collaboration.
- Implemented frontend dashboards and backend API modules focused on privacy, data integrity, and workflow usability.

SenseTime Group Limited, Software Engineer

2020.11 – 2023.9

Tech: TypeScript, React, Electron, NodeJS, Webpack, CI/CD

- Awarded **CIT Team Best Employee (2021)** for outstanding engineering impact and delivery.
- Led frontend development across shared component libraries, internal development tooling, and enterprise platforms.
- Built and maintained group-wide OA systems and Enterprise WeChat automation modules.
- Developed AR navigation and enterprise platforms used across multiple business units.

6 PUBLICATION

6G-enabled Edge AI for Metaverse: Challenges, Methods, and Future Research Directions. JCN 2022. **Authors:** Luyi Chang, Zhe Zhang, Pei Li, Shan Xi, Wei Guo, **Yukang Shen**, Zehui Xiong, Jiawen Kang, Dusit Niyato, Xiuquan Qiao, Yi Wu.

7 SKILLS

Programming: Python, TypeScript, Node.js, React, Electron, PyTorch

AI: Computer Vision, 3D Vision, SLAM, Multimodal Perception **Systems:** Synthetic Data Pipelines, Dataset Tooling